Grothendieck's definition of a group

Terence Tao · 19 October 2009

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In his wonderful article "On proof and progress in mathematics", Bill Thurston describes (among many other topics) how one's understanding of given concept in mathematics (such as that of the derivative) can be vastly enriched by viewing it simultaneously from many subtly different perspectives; in the case of the derivative, he gives seven standard such perspectives (infinitesimal, symbolic, logical, geometric, rate, approximation, microscopic) and then mentions a much later perspective in the sequence (as describing a flat connection for a graph).

One can of course do something similar for many other fundamental notions in mathematics. For instance, the notion of a group G can be thought of in a number of (closely related) ways, such as the following:

- (0) Motivating examples: A group is an abstraction of the operations of addition/subtraction or multiplication/division in arithmetic or linear algebra, or of composition/inversion of transformations.
- (1) Universal algebraic: A group is a set G with an identity element e, a unary inverse operation $\cdot^{-1}: G \to G$, and a binary multiplication operation $\cdot : G \times G \to G$ obeying the relations (or axioms) $e \cdot x = x \cdot e = x$, $x \cdot x^{-1} = x^{-1} \cdot x = e$, $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in G$.
- (2) Symmetric: A group is all the ways in which one can transform a space V to itself while preserving some object or structure O on this space.
- (3) Representation theoretic: A group is identifiable with a collection of transformations on a space V which is closed under composition and inverse, and contains the identity transformation.
- (4) Presentation theoretic: A group can be generated by a collection of generators subject to some number of relations.
- (5) Topological: A group is the fundamental group $\pi_1(X)$ of a connected topological space X.
- (6) Dynamic: A group represents the passage of time (or of some other variable(s) of motion or action) on a (reversible) dynamical system.
- (7) Category theoretic: A group is a category with one object, in which all morphisms have inverses.
- (8) Quantum: A group is the classical limit $q \to 0$ of a quantum group.
- etc.

One can view a large part of group theory (and related subjects, such as representation theory) as exploring the interconnections between various of these perspectives. As one's understanding of the subject matures, many of these formerly distinct perspectives slowly merge into a single unified perspective.

From a recent talk by Ezra Getzler, I learned a more sophisticated perspective on a group, somewhat analogous to Thurston's example of a sophisticated perspective on a derivative (and coincidentally, flat connections play a central role in both):

- (37) Sheaf theoretic: A group is identifiable with a (set-valued) sheaf on the category of simplicial complexes such that the morphisms associated to collapses of d-simplices are bijective for $d > 1$ (and merely surjective for $d \leq 1$).

This interpretation of the group concept is apparently due to Grothendieck, though it is motivated also by homotopy theory. One of the key advantages of this interpretation is that it generalises easily to the notion of an n-group (simply by replacing 1 with n in (37)), whereas the other interpretations listed earlier require a certain amount of subtlety in order to generalise correctly (in particular, they usually themselves require higher-order notions, such as n-categories).

The connection of (37) with any of the other perspectives of a group is elementary, but not immediately obvious; I enjoyed working out exactly what the connection was, and thought it might be of interest to some readers here, so I reproduce it below the fold.

[Note: my reconstruction of Grothendieck's perspective, and of the appropriate terminology, is likely to be somewhat inaccurate in places: corrections are of course very welcome.]

— 1. Flat connections —

To see the relationship between (37) and more traditional concepts of a group, such as (1), we will begin by recalling the machinery of flat connections.

Let G be a group, X be a topological space. A principal G-connection ω on X can be thought of as an assignment of a group element $\omega(\gamma) \in G$ to every path γ in X which obey the following four properties:

- Invariance under reparameterisation: if γ' is a reparameterisation of γ, then $\omega(\gamma) = \omega(\gamma')$.
- Identity: If γ is a constant path, then $\omega(\gamma)$ is the identity element.
- Inverse: If $-\gamma$ is the reversal of a path γ, then $\omega(-\gamma)$ is the inverse of $\omega(\gamma)$.
- Groupoid homomorphism: If γ_2 starts where γ_1 ends (so that one can define the concatenation $\gamma_1 + \gamma_2$), then $\omega(\gamma_1 + \gamma_2) = \omega(\gamma_2)\omega(\gamma_1)$. (Depending on one's conventions, one may wish to reverse the order of the group multiplication on the right-hand side.)

Intuitively, $\omega(\gamma)$ represents a way to use the group G to connect (or "parallel transport") the fibre at the initial point of γ to the fibre at the final point; see this previous blog post for more discussion. Note that the identity property is redundant, being implied by the other three properties.

We say that a connection ω is *flat* if $\omega(\gamma)$ is the identity element for every "short" closed loop γ, thus strengthening the identity property. One could define "short" rigorously (e.g. one could use "contractible" as a substitute), but we will prefer here to leave the concept intentionally vague.

Typically, one studies connections when the structure group G and the base space X are continuous rather than discrete. However, there is a combinatorial model for connections which is suitable for discrete groups, in which the base space X is now an (abstract) simplicial complex Δ – a vertex set V, together with a number of *simplices* in V, by which we mean ordered $d+1$-tuples $(x_0, ..., x_d)$ of distinct vertices in V for various integers d (with d being the *dimension* of the simplex $(x_0, ..., x_d)$). In our definition of a simplicial complex, we add the requirement that if a simplex lies in the complex, then all faces of that simplex (formed by removing one of the vertices, but leaving the order of the remaining vertices unchanged) also lie in the complex. We also assume a well defined *orientation*, in the sense that every $d+1$-tuple $\{x_0, ..., x_d\}$ is represented by at most one simplex (thus, for instance, a complex cannot contain both an edge $(0,1)$ and its reversal $(1,0)$). Though it will not matter too much here, one can think of the vertex set V here as being restricted to be finite.

A *path* γ in a simplicial complex Δ is then a sequence of 1-simplices (x_i, x_{i+1}) or their formal reverses $-(x_i, x_{i+1})$, with the final point of each 1-simplex being the initial point of the next. If G is a (discrete) group, a *principal G-connection* ω on Δ is then an assignment of a group element $\omega(\gamma) \in G$ to each such path γ, obeying the groupoid homomorphism property and the inverse property (and hence the identity property). Note that the reparameterisation property is no longer needed in this abstract combinatorial model. Note that a connection can be determined by the group elements $\omega(b \leftarrow a)$ it assigns to each 1-simplex (a, b). (I have written the simplex $b \leftarrow a$ from right to left, as this makes the composition law cleaner.)

So far, only the 1-skeleton (i.e. the simplices of dimension at most 1) of the complex have been used. But one can use the 2-skeleton to define the notion of a *flat* connection: we say that a principal G-connection ω on Δ is flat if the boundary of every 2-simplex (a, b, c), oriented appropriately, is assigned the identity element, or more precisely that $\omega(c \leftarrow a)^{-1}\omega(c \leftarrow b)\omega(b \leftarrow a) = e$, or in other words that $\omega(c \leftarrow a) = \omega(c \leftarrow b)\omega(b \leftarrow a)$; thus, in this context, a "short loop" means a loop that is the boundary

of a 2-simplex. Note that this corresponds closely to the topological concept of a flat connection when applied to, say, a triangulated manifold.

Fix a group G. Given any simplicial complex Δ, let $\mathcal{O}(\Delta)$ be the set of flat connections on Δ. One can get some feeling for this set by considering some basic examples:

- If Δ is a single 0-dimensional simplex (i.e. a point), then there is only the trivial path, which must be assigned the identity element e of the group. Thus, in this case, $\mathcal{O}(\Delta)$ can be identified with $\{e\}$.
- If Δ is a 1-dimensional simplex, say $(0,1)$, then the path from 0 to 1 can be assigned an arbitrary group element $\omega(1 \leftarrow 0) \in G$, and this is the only degree of freedom in the connection. So in this case, $\mathcal{O}(\Delta)$ can be identified with G.
- Now suppose Δ is a 2-dimensional simplex, say $(0,1,2)$. Then the group elements $\omega(1 \leftarrow 0)$ and $\omega(2 \leftarrow 1)$ are arbitrary elements of G, but $\omega(2 \leftarrow 0)$ is constrained to equal $\omega(2 \leftarrow 1)\omega(1 \leftarrow 0)$. This determines the entire flat connection, so $\mathcal{O}(\Delta)$ can be identified with G^2.
- Generalising this example, if Δ is a k-dimensional simplex, then $\mathcal{O}(\Delta)$ can be identified with G^k.

An important operation one can do on flat connections is that of *pullback*. Let $\phi : \Delta \to \Delta'$ be a *morphism* from one simplicial complex Δ to another Δ'; by this, we mean a map from the vertex set of Δ to the vertex set of Δ' such that every simplex in Δ maps to a simplex in Δ' in an order preserving manner (though note that ϕ is allowed to be non-injective, so that the dimension of the simplex can decrease by mapping adjacent vertices to the same vertex). Given such a morphism, and given a flat connection ω' on Δ', one can then pull back that connection to yield a flat connection $\phi^*\omega'$ on Δ, defined by the formula

$$\phi^*\omega'(w \leftarrow v) := \omega'(\phi(w) \leftarrow \phi(v))$$

for any 1-simplex (v,w) in Δ, with the convention that $\omega'(u \leftarrow u)$ is the identity for any u. It is easy to see that this is still a flat connection. Also, if $\phi : \Delta \to \Delta'$ and $\psi : \Delta' \to \Delta''$ are morphisms, then the operations of pullback by ψ and then by ϕ compose to equal the operation of pullback by $\psi \circ \phi$: $\phi^*\psi^* = (\psi \circ \phi)^*$. In the language of category theory, pullback is a contravariant functor from the category of simplicial complexes to the category of sets (with each simplicial complex being mapped to its set of flat connections).

A special case of a morphism is an *inclusion morphism* $\iota : \Delta \to \Delta'$ to a simplicial complex Δ' from a subcomplex Δ. The associated pullback operation is the *restriction* operation, that maps a flat connection ω' on Δ' to its restriction $\omega' \downharpoonright_{\Delta}$ to Δ.

— 2. Sheaves —

We currently have a set $\mathcal{O}(\Delta)$ of flat connections assigned to each simplicial complex Δ, together with pullback maps (and in particular, restriction maps) connecting these sets to each other. One can easily observe that this system of structures obeys the following axioms:

- (Identity) There is only one flat connection on a point.
- (Locality) If $\Delta = \Delta_1 \cup \Delta_2$ is the union of two simplicial complexes, then a flat connection on Δ is determined by its restrictions to Δ_1 and Δ_2. In other words, the map $\omega \mapsto (\omega \downharpoonright_{\Delta_1}, \omega \downharpoonright_{\Delta_2})$ is an injection from $\mathcal{O}(\Delta)$ to $\mathcal{O}(\Delta_1) \times \mathcal{O}(\Delta_2)$.
- (Gluing) If $\Delta = \Delta_1 \cup \Delta_2$, and ω_1, ω_2 are flat connections on Δ_1, Δ_2 which agree when restricted to $\Delta_1 \cap \Delta_2$, (and if the orientations of Δ_1, Δ_2 on the intersection $\Delta_1 \cap \Delta_2$ agree) then there exists a flat connection ω on Δ which agrees with ω_1, ω_2 on Δ_1, Δ_2. (Note that this gluing of ω_1 and ω_2 is unique, by the previous axiom. It is important that the orientations match; we cannot glue $(0,1)$ to $(1,0)$, for instance.)

One can consider more abstract assignments of sets to simplicial complexes, together with pullback maps, which obey these three axioms. A system which obeys the first two axioms is known as a *pre-sheaf*, while a system that obeys all three is known as a sheaf. (One can also consider pre-sheaves and

sheaves on more general topological spaces than simplicial complexes, for instance the spaces of smooth (or continuous, or holomorphic, etc.) functions (or forms, sections, etc.) on open subsets of a manifold form a sheaf.)

Thus, flat connections associated to a group G form a sheaf. But flat connections form a special type of sheaf that obeys an additional property (listed above as (37)). To explain this property, we first consider a key example when $\Delta = (0,1,2)$ is the standard 2-simplex (together with subsimplices), and Δ' is the subcomplex formed by removing the 2-face $(0,1,2)$ and the 1-face $(0,2)$, leaving only the 1-faces $(0,1), (1,2)$ and the 0-faces $0, 1, 2$. Then of course every flat connection on Δ restricts to a flat connection on Δ'. But the flatness property ensures that this restriction is invertible: given a flat connection on Δ', there exists a unique extension of this connection back to Δ. This is nothing more than the property, remarked earlier, that to specify a flat connection on the 2-simplex $(0,1,2)$, it suffices to know what the connection is doing on $(0,1)$ and $(1,2)$, as the behaviour on the remaining edge can then be deduced from the group law; conversely, any specification of the connection on those two 1-simplices determines a connection on the remainder of the 2-simplex.

This observation can be generalised. Given any simplicial complex Δ, define a k-dimensional *collapse* Δ' of Δ to be a simplicial complex obtained from Δ by removing the interior of a k-simplex, together with one of its faces; thus for instance the complex consisting of $(0,1), (1,2)$ (and subsimplices) is a 2-dimensional collapse of the 2-simplex $(0,1,2)$ (and subsimplices). We then see that the sheaf of flat connections obeys an additional axiom:

- (Grothendieck's axiom) If Δ' is a k-dimensional collapse of Δ, then the restriction map from $\mathcal{O}(\Delta)$ to $\mathcal{O}(\Delta')$ is surjective for all k, and bijective for $k \geq 2$.

This axiom is trivial for $k = 0$. For $k = 1$, it is true because if an edge (and one of its vertices) can be removed from a complex, then it is not the boundary of any 2-simplex, and the value of a flat connection on that edge is thus completely unconstrained. (In any event, the $k = 1$ case of this axiom can be deduced from the sheaf axioms.) For $k = 2$, it follows because if one can remove a 2-simplex and one of its edges from a complex, then the edge is not the boundary of any other 2-simplex and thus the connection on that edge only constrained to precisely be the product of the connection on the other two edges of the 2-simplex. For $k = 3$, it follows because if oen removes a 3-simplex and one of its 2-simplex faces, the constraint associated to that 2-simplex is implied by the constraints coming from the other three faces of the 3-simplex (I recommend drawing a tetrahedron and chasing some loops around to see this), and so one retains bijectivity. For $k \geq 4$, the axiom becomes trivial again because the k-simplices and $k-1$ -simplices have no impact on the definition of a flat connection.

Grothendieck's beautiful observation is that the converse holds: if a (concrete) sheaf $\Delta \mapsto \mathcal{O}(\Delta)$ obeys Grothendieck's axiom, then it is equivalent to the sheaf of flat connections of some group G defined canonically from the sheaf. Let's see how this works. Suppose we have a sheaf $\Delta \mapsto \mathcal{O}(\Delta)$, which is concrete in the sense that $\mathcal{O}(\Delta)$ is a set, and the morphisms between these sets are given by functions. In analogy with the preceding discussion, we'll refer to elements of $\mathcal{O}(\Delta)$ as (abstract) flat connections, though *a priori* we do not assume there is a group structure behind these connections.

By the sheaf axioms, there is only one flat connection on a point, which we will call the trivial connection. Now consider the space $\mathcal{O}(0,1)$ flat connections on the standard 1-simplex $(0,1)$. If the sheaf was indeed the sheaf of flat connections on a group G, then $\mathcal{O}(0,1)$ is canonically identifiable with G. Inspired by this, we will *define* G to equal the space $\mathcal{O}(0,1)$ of flat connections on $(0,1)$. The flat connections on any other 1-simplex (u,v) can then be placed in one-to-one correspondence with elements of G by the morphism $u \mapsto 0, v \mapsto 1$, so flat connections on (u,v) can be viewed as being *equivalent* to an element of G.

At present, G is merely a set, not a group. To make it into a group, we need to introduce an identity element, an inverse operation, and a multiplication operation, and verify the group axioms.

To obtain an identity element, we look at the morphism from $(0,1)$ to a point, and pull back the trivial connection on that point to obtain a flat connection e on $(0,1)$, which we will declare to be the identity element. (Note from the functorial nature of pullback that it does not matter which point we choose for this.)

Now we define the multiplication operation. Let $g, h \in G$, then g and h are flat connections on $(0,1)$. By using the morphism $i \mapsto i-1$ from $(1,2)$ to $(0,1)$, we can pull back h to $(1,2)$ to create a flat connection $\tilde{h}$ on $(1,2)$ that is equivalent to h. The restriction of g and $\tilde{h}$ to the point 1 is trivial, so by the gluing axiom we can glue g and $\tilde{h}$ to a flat connection on the complex $(0,1), (1,2)$. By Grothendieck's axiom, one can then uniquely extend this connection to the 2-simplex $(0,1,2)$, which can then be restricted to the edge $(0,2)$. Mapping this edge back to $(0,1)$, we obtain an element of G, which we will define to be hg.

This operation is closed. To verify the identity property, observe that if $g \in G$, then by starting with the simplex $(0,1,2)$ and pulling back g under the morphism that sends 2 to 1 but is the identity on $0, 1$, we obtain a flat connection on $(0,1,2)$ which is equal to g on $(0,1)$, equivalent to the identity on $(1,2)$, and is equivalent to g on $(0,2)$ (after identifying $(0,2)$ with $(0,1)$). From the definition of group multiplication, this shows that $eg = g$; a similar argument (using a slightly different morphism from $(0,1,2)$ to $(0,1)$) gives $ge = g$.

Now we establish associativity. Let $f, g, h \in G$. Using the definition of multiplication, we can create a flat connection on the 2-simplex $(0,1,2)$ which equals h on $(0,1)$, is equivalent to g on $(1,2)$, and is equivalent to gh on $(0,2)$. We then glue on the edge $(2,3)$ and extend the flat connection to be equivalent to f on $(2,3)$. Using Grothendieck's axiom and the definition of multiplication, we can then extend the flat connection to the 2-simplex $(0,2,3)$ to be equivalent to $f(gh)$ on $(0,3)$. By another use of that axiom, we can also extend the flat connection to the 2-simplex $(1,2,3)$, to be equivalent to fg on $(1,3)$. Now that we have three of the four faces of the 3-simplex $(0,1,2,3)$, we can now apply the $k=3$ case of Grothendieck's axiom and extend the flat connection to the entire 3-simplex, and in particular to the 2-simplex $(0,1,3)$. Using the definition of multiplication again, we thus see that $f(gh) = (fg)h$, giving associativity.

Next, we establish the inverse property. It will suffice to establish the existence of a left-inverse and a right-inverse for each group element, since the associativity property will then guarantee that these two inverses equal each other. We shall just establish the left-inverse property, as the right-inverse is analogous. Let $g \in G$ be arbitrary. By the gluing axiom, one can form a flat connection on the complex $(0,1), (0,2)$ which equals g on $(0,1)$ and is equivalent to the identity on $(0,2)$. By Grothendieck's axiom, this can be extended to a flat connection on $(0,1,2)$; the restriction of this connection to $(1,2)$ is equivalent to some element of G, which we define to be g^{-1}. By construction, $g^{-1}g = e$ as required.

We have just shown that G is a group. The last thing to do is to show that this abstract sheaf $\mathcal{O}$ can be indeed identified with the sheaf of G-flat connections. This is fairly straightforward: given an abstract flat connection on a complex, the restriction of that connection to any edge is equivalent to an element of G. To verify that this genuinely determines a G-connection on that complex, we need to verify that if (u,v) and (v,u) are both in the complex, that the group elements g, h assigned to these edges invert each other. But we can pullback $(u,v), (v,u)$ to the 2-simplex $(0,1,2)$ by mapping $0, 2$ to u and 1 to v, creating a flat connection that is equal to g on $(0,1)$, equivalent to h on $(1,2)$, and equivalent to the identity on $(0,2)$; by definition of multiplication or inverse we conclude that g, h invert each other as desired.

Thus the abstract connection defines a G-connection. From the definition of multiplication it is also clear that every 2-simplex in the complex imposes the right relation on the three elements of G associated to the edges of that simplex that makes the G-connection flat. Thus we have a canonical way to associate a G-flat connection to each abstract flat connection; the only remaining things to do are verify that this association is bijective.

We induct on the size of the complex. If the complex is not a single simplex, the claim follows from the induction hypothesis by viewing the complex as the union of two (possibly overlapping) smaller complexes, and using the gluing and locality axioms. So we may assume that the complex consists of a single simplex. If the simplex is 0 or 1-dimensional the claim is easy; for $k \geq 2$ the claim follows from Grothendieck's axiom (which applies both for the abstract flat connections (by hypothesis) and for G-flat connections (as verified earlier)) and the induction hypothesis.

Comments (25)

 **Anonymous** · 19 October 2009

Wonder if this post is missing an expository tag. *[Added, thanks. (It seems my readers are more familiar with my organisational system than I am.) – T.]*

 **Lior** · 19 October 2009

In the definition of a flat connection, γ_2 should begin where γ_1 *ends.*
[Corrected, thanks – T.]

>  **Lior** · 19 October 2009
>
> You also have "At present, G is merely a group, not a set" rather than the reverse. *[Corrected, thanks – T.]*

 **anonymous** · 19 October 2009

In the interpretation (37) the wording is a little bit garbled: "such that the morphisms associated to collapse of {d}-simplices for are bijective" should probably be "such that the morphisms associated to collapses of {d}-simplices are bijective". *[Fixed, thanks – T.]*

 **timur** · 20 October 2009

Does numbering (37) mean you had a list up to (36) before coming across this interpretation?

>  **Terence Tao** · 20 October 2009
>
> Well, no, but certainly there are more ways to view a group than are listed here. The precise number (37) is also used in Thurston's article, referenced above.

 **A semana nos arXivs... « Ars Physica** · 20 October 2009

[...] Grothendieck's definition of a group (alguém disse Langlands Duality?) [...]

 **Raj** · 20 October 2009

I got stuck for a little bit on this: "pullback is a contravariant functor from the category of simplicial complexes to the category of flat connections." I think what you mean to say is that pullback gives us a contravariant functor from the category of simplicial complexes to the category of sets, where at the level of objects a simplicial complex gets sent to the set of all flat connections on it.
Is my interpretation right, or am I missing something?
[Yes, thanks, this is what was intended. I've corrected the text. -T]

 **Raj** · 20 October 2009

Also, this reminded me of another point of view which is closely related to the category-theoretic definition (7). Namely, if you think of a group as a category with one object and invertible morphisms, then you can take its nerve, which is a simplicial complex with only one 0-simplex. This simplicial complex is special in two other ways: a) it is Kan, which means that any (k-1)-dimensional horn can be filled by a k-simplex, and b) this horn-filling is unique for k > 1. Another way to state these conditions is by defining various "horn maps" that take k-simplices to (k-1)-dimensional horns. Then the Kan property says that these horn maps are bijective, and the uniqueness property says that the maps are bijective for k > 1.
Superficially, this looks very similar to Grothendieck's axiom, and I'm guessing it's not just a coincidence. Do you know how to relate the two?

 **Raj** · 20 October 2009

sorry, I meant to say "the Kan property says that these horn maps are surjective".

>  **Terence Tao** · 20 October 2009
>
> One would have to ask a category theorist for a proper answer, but I do recall Ezra mentioning a close connection of Grothendieck's interpretation of a group with nerves in his talk. Unfortunately I do not recall the precise nature of the connection…

Top Posts « WordPress.com · 21 October 2009

[…] Grothendieck's definition of a group In his wonderful article "On proof and progress in mathematics", Bill Thurston describes (among many other […] […]

Tales from the Tubes—23/10/09 | Young Australian Skeptics · 22 October 2009

[…] via Terence Tao, a good (but older) article on understanding and communicating […]

 **Ben Wieland** · 26 October 2009

I think Raj is right that these two definitions are pretty much the same. You can replace simplicial complexes with pretty much any geometric category, like simplicial sets. The n-simplex is not symmetric in the category of simplicial sets, so you have several different horn conditions, rather than the single one for simplicial sets. If you only impose some of them, you can expand the definition to monoids, dropping inverses.

The Kan complex version doesn't mention sheaves, because a sheaf on the category of simplicial sets is the same thing as a simplicial set. This is a general principle that a sheaf on the category of sheaves is represented by a sheaf in the original category of sheaves. In particular, if we have no sheaf condition, that is, if we take a diagram (category) D and look at all functors from D to sets, as in the case of simplicial sets, a sheaf on this category is determined by its restriction to the sheaves represented by objects of D, and thus it essentially a functor from D to sets.

In fact, the category of sheaves on the category of simplicial complexes is not a lot larger than the category of simplicial complexes. By locality, a sheaf is determined by its restriction to the full subcategory of standard simplices. I claim that any functor yields a sheaf, just like a simplicial set is any functor on the full subcategory (of simplicial sets) spanned by the standard simplices.

The category of functors here is a bit bigger than the category of simplicial complexes, but not a lot bigger. It just adds all colimits which contains things like an interval with the two ends glued together, or two intervals with the same pair of endpoints. Depending on your definition of morphism of simplicial complex, objects like these are all the extra objects, that is, simplices glued along faces of the same dimension. (whereas simplicial sets are simplices glued along linear maps more general than inclusions)

>  **Ben Wieland** · 27 October 2009
>
> In my first paragraph, that should have been "the single [horn condition] for simplicial *complexes*," in contrast to several for simplicial *sets*.
>
> In my last paragraph, I imply that simplicial sets allow more general gluing maps than simplicial complexes. That is not true: neither condition is weaker. What is true is that simplical sets allow gluing along non-injective maps, such as gluing the boundary of a 2-simplex to a point to get a 2-sphere (with a single top cell). But simplicial sets are restricted in their morphisms and gluing maps by the order of the vertices of each simplex.

 **Grothendieck, Sistemas Dinâmicos, Dualidade de Langlands e Higgs Bundles… « Ars Physica** · 31 October 2009

[…] Grothendieck's definition of a group, […]

 **Nonconventional ergodic averages and multiple recurrence for von Neumann dynamical systems « What's new** · 29 December 2009

[...] inspired by Grothendieck's interpretation of a group as a sheaf of flat connections, which I blogged about recently, and which I will discuss below the [...]

 **LMSchmitt** · 6 July 2010

It should, perhaps, be noticed that a set G with associative operation o and left-neutral element 1=c , i.e., 1 o x =x, as-well-as left inverse I(x) in G for every x in G, i.e., I(x) o x = 1, is already a group in the above-mentioned universal-algebra sense.

 **Cayley graphs and the geometry of groups « What's new** · 10 July 2010

[...] (giving rise to Lie groups) or a topological space (giving rise to topological groups). (See also this post for a number of other ways to think about [...]

Grothendieck 1928-2014 — genius or crank? BOTH! | Turing Machine · 26 November 2014

[...] 12. Grothendieck's definition of a group | What's new [...]

 **vznvzn** · 26 November 2014

saw your comment on credit to grothendieck on gowers blog & also found this blog via reddit or surfing.(?) here is some tribute to grothendieck with some tie-in to gowers essay "the two cultures of mathematics". many links, top obituaries etc., and (full disclosure) a so-called "crank" angle, also cites this page.

Grothendieck 1928-2014 — Genius or Crank? BOTH. hope/ encourage you to comment/ blog on grothendieck at some pt!

 **ashwin1729** · 3 December 2014

I found this book,Visual Group Theory by Nathan Carter absolutely amazing.

Grothendieck – A group as a sheaf over simplicial complexes | TRIFORCE STUDIO · 5 December 2015

[...] this blog post, Terence Tao gives the following definition of a [...]

Multiple ways of understanding | The Lumber Room · 26 March 2016

[...] has also been best extended by Terence Tao, who in an October 2009 blog post on Grothendieck's definition of a group gave several ways of thinking about a [...]

How a group represents the passage of time? BootMath · 6 April 2018

[...] Grothendieck's definition of a group [...]